INVERTIS UNIVERSITY BAREILLY
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BCS-201- Programming For Problem Solving

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Course Outcomes:

The course will enable the students

- To formulate simple algorithms for arithmetic and logical problems
- To translate the algorithms to programs (in C language)
- To test and execute the programs and correct syntax and logical errors
- To implement conditional branching, iteration and recursion
- To decompose a problem into functions and synthesize a complete program using divide and conquer approach
- To use arrays, pointers and structures to formulate algorithms and programs
- To apply programming to solve matrix addition and multiplication problems and searching and sorting problems
- To apply programming to solve simple numerical method problems, namely rot finding of function, differentiation of function and simple integration

MODULE-1

Programming language and environment: Introduction, Types of Languages and their Advantages and Disadvantages, Comparison, language translators. Programming Environment, Format of C program, Compiling and linking, executing a C Program, Write and execute the first program.
Overview of 'C'- Introduction and history, characteristics, Application, Program development life cycle.
Data Types and Storage Classes: Introduction, Types of data type, Data type modifier, Size and ranges of data types, Tokens, Variables. Storage classes- introduction, types and features.

MODULE-2

Operators and Expressions: Introduction, Types based on operands used, Precedence and Associativity, Arithmetic Expression, Evaluation of Expression, Type Conversions in Expressions, Mathematical Functions.

Decision Making and Branching: Introduction, Statement and Blocks**,** Conditional Statements **(**if statement**,** Switch statement**,** Ternary operator**,** goto statement), The break and continue statement.
Program Loops and Iteration: Introduction, Types, Uses of while, do and for loops, multiple loop variables, assignment operators.
Modular Programming: Introduction, advantages, Functions and their types, Declaration and Definition of a function, Scope rules and global variables, separate compilation, and linkage, building your own modules, Call by value and Call by reference.

MODULE-3

Arrays and Strings: Introduction, Array notation and representation, Declaring and initializing arrays, Manipulating array elements, using multidimensional arrays, String handling functions**.**
Structures: Purpose and usage of structures, declaring and defining structure, Assigning Values to variable, Initializing Structures, Arrays of Structures, Structures and Functions

Pointers to Objects: Introduction, Void and Null Pointers, Pointer and address arithmetic, pointer operations

and declarations, using pointers as function arguments, Pointer and Structure.

Basic Concepts of Dynamic Memory Allocation, Searching and Sorting, C Preprocessor directives and their types.

File Handling: Introduction, Types of Files, Defining, Opening and Closing a File, Input/output Operations on Files, Some Basic Functions

Text Books:

1. *Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill*
2. *E. Balaguruswamy, Programming in ANSI C, Tata McGraw-Hill*

Reference Books:

1. *Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India*
2. *Jeri R. Hanly, Elliot B. Koffman, "Problem Solving and Program Design in C", Pearson Addison-Wesley,2006.*
3. *Behrouz A. Forouzan, Richard F. Gilberg,Computer Science- "A Structured Programming Approach Using C", Thomson, Third Edition [India Edition],2007.*
4. *Victor Alvarado, Mocygo San Jose,"M. S. Office For ME Word, Excel, Power Point, CA"*
5. *Yashwant Kanetker, "Let us C", BPB Publication,2008.*
6. *Detiel & Detiel, "„C" How to program, ISBN: 0132404168, 5th Edition,2007".*